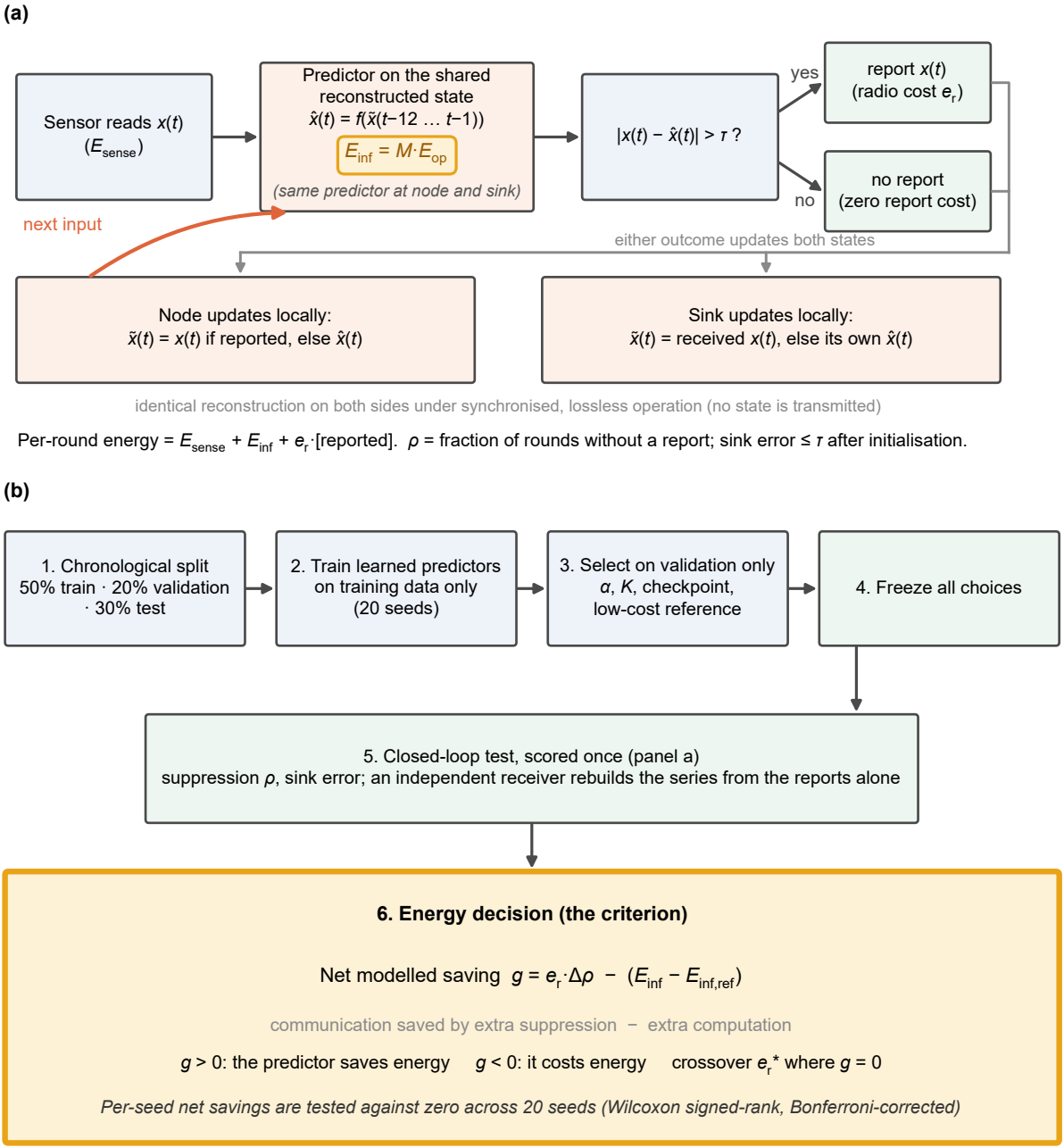

(a)
Sensor reads x(t)
(E_sense)
Predictor on the shared reconstructed state
x̂(t) = f(x̃(t−12 … t−1))
E_inf = M·E_op
(same predictor at node and sink)
|x(t) − x̂(t)| > τ ?
yes
report x(t)
(radio cost e_r)
no
no report
(zero report cost)
next input
either outcome updates both states
Node updates locally:
x̃(t) = x(t) if reported, else x̂(t)
Sink updates locally:
x̃(t) = received x(t), else its own x̂(t)
identical reconstruction on both sides under synchronised, lossless operation (no state is transmitted)
Per-round energy = E_sense + E_inf + e_r·[reported]. ρ = fraction of rounds without a report; sink error ≤ τ after initialisation.
(b)
1. Chronological split
50% train · 20% validation
· 30% test
2. Train learned predictors
on training data only
(20 seeds)
3. Select on validation only
α, K, checkpoint,
low-cost reference
4. Freeze all choices
5. Closed-loop test, scored once (panel a)
suppression ρ, sink error; an independent receiver rebuilds the series from the reports alone
6. Energy decision (the criterion)
Net modelled saving g = e_r·Δρ − (E_inf − E_inf,ref)
communication saved by extra suppression − extra computation
g > 0: the predictor saves energy
g < 0: it costs energy
crossover e_r* where g = 0
Per-seed net savings are tested against zero across 20 seeds (Wilcoxon signed-rank, Bonferroni-corrected)